Laboratory Exercise

Normalizing Student Flat Data Using Excel Power Query

Objective

In this laboratory exercise, you will normalize a flat student record dataset using Microsoft Excel Power Query. The goal is to transform unstructured data into properly organized tables that follow the Relational Data Model and reduce data redundancy.

Materials

- Microsoft Excel (with Power Query)
- Provided Excel file: **Student Flat Data**

Background

The provided dataset contains student records stored in a single flat table. This structure causes redundancy and violates normalization principles. Using Power Query, you will separate the data into multiple related tables representing different entities.

Procedure:

Part 1. Perform Data Cleaning and Preparation

1. Check the column profile and remove null entries if there's any
2. Make sure all the columns are assigned with the correct data types/format (Numeric for number data, Date and Text)
3. You may Split **Last Name** and **First Name** of the Student Name column
4. Add a Column that will Display **"Freshmen, Sophomore, Junior and Senior"** in relation to Year Level – Rename the new Column as **Year Label**
5. Make sure all columns are given proper descriptive names
6. Split the course code into 2 columns **Course Code – Course Number**
7. Affix and or replace all @school.edu into cca.edu.ph in the Email column assuming all students are cca students
8. Split the schedule column into "Days" "Time Start" and "Time End"
9. Make sure there are NO Duplicate Data. Check Column Profile

Part 2. Normalization and Creating the Data Model

1. Examine the dataset and identify columns that belong to different entities, such as: o

   Student information

   o Course information
   o Department information

2. In the Power Query Editor, review all columns and determine which attributes belong to each entity.

3. Right-click the original query and select **Duplicate** to create a working copy for each entity.

Creating the Students Table

7. Rename one duplicated query to **Students**.

8. Keep only student-related attributes such as:

    o Student ID

    o First Name

    o Last Name

    o Gender

    o Birthdate

    o Contact Number

    o Email

    o Course ID

9. Remove all unrelated columns.

10. Refer to the original dataset to identify the Columns that you will include

11. Use **Remove Duplicates** to ensure that each Student ID appears only once.

Creating the Courses Table

11. Rename another duplicated query to **Courses**.

12. Keep only course-related attributes such as:

• Course ID

• Course Name

• Department ID

13. Remove duplicate records so that each course appears only once.

Creating the Departments Table

14. Rename another duplicated query to **Departments**.

15. Keep only department-related attributes such as:

• Department ID

• Department Name

16. Remove duplicate rows to ensure unique department records.

Finalizing the Data

17. Verify that each table has a clear primary key:

• Students: Student ID

• Courses: Course ID

• Departments: Department ID

18. Ensure foreign keys are properly placed:

• Course ID in the Students table

• Department ID in the Courses table

19. Rename all queries clearly and consistently.

20. Click **Close & Load to** load each table into Excel as a separate worksheet.

Output Requirements

• **Three normalized tables:**

  o Students

  o Courses

  o Departments

• No duplicate records

• Clearly defined primary and foreign keys

• Clean and readable data structure

Midterm Lab Task 2. Data Preparation and Modeling using Power Query

CLEAN DATA:

Queries [6]

- Student_Flat_Data
- practice
- stud_dim
- course_dim
- dept_dim
- info_Fact

```
= Table.RenameColumns(#"Removed Columns2",{{"course_dim.CourseCode", "CourseCode"}, {"dept_dim.dept_id", "dept_id"}})
```

	StudentID	First Name	Last Name	Email	dept_id	YearLevel	Year Label	CourseCode	Instructor
1	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	CS204	Mr. Cruz
2	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	IT203	Mr. Santos
3	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	IT203	Mr. Cruz
4	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	IT203	Mr. Cruz
5	S004	Dan	Santos	dan.santos@cca.edu.ph	D-102	3	Junior and Senior	IT101	Ms. Lim
6	S004	Dan	Santos	dan.santos@cca.edu.ph	D-102	3	Junior and Senior	CS102	Ms. Lim
7	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	IT101	Ms. Lim
8	S004	Dan	Santos	dan.santos@cca.edu.ph	D-102	3	Junior and Senior	CS102	Mr. Cruz
9	S003	Cara	Lim	cara.lim@cca.edu.ph	D-102	1	Freshmen	CS204	Mr. Cruz
10	S003	Cara	Lim	cara.lim@cca.edu.ph	D-102	1	Freshmen	IT203	Mr. Santos
11	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	CS204	Mr. Cruz
12	S004	Dan	Santos	dan.santos@cca.edu.ph	D-102	3	Junior and Senior	CS102	Ms. Lim
13	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	CS102	Ms. Lim
14	S005	Eva	Flores	eva.flores@cca.edu.ph	D-101	2	Sophomore	CS204	Ms. Lim
15	S005	Eva	Flores	eva.flores@cca.edu.ph	D-101	2	Sophomore	IT305	Ms. Lim
16	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	CS102	Mr. Cruz
17	S003	Cara	Lim	cara.lim@cca.edu.ph	D-102	1	Freshmen	IT101	Mr. Santos
18	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	IT101	Ms. Lim
19	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	CS204	Mr. Santos
20	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	IT101	Mr. Santos
21	S005	Eva	Flores	eva.flores@cca.edu.ph	D-101	2	Sophomore	IT305	Mr. Cruz
22	S005	Eva	Flores	eva.flores@cca.edu.ph	D-101	2	Sophomore	IT305	Ms. Lim
23	S005	Eva	Flores	eva.flores@cca.edu.ph	D-101	2	Sophomore	CS102	Mr. Cruz
24	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	CS204	Ms. Lim
25	S002	Ben	Reyes	ben.reyes@cca.edu.ph	D-101	2	Sophomore	IT305	Mr. Cruz
26	S003	Cara	Lim	cara.lim@cca.edu.ph	D-102	1	Freshmen	CS102	Mr. Santos
27	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	CS204	Mr. Santos
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35	S005	Eva	Flores	eva.flores@cca.edu.ph	D-101	2	Sophomore	IT101	Mr. Cruz
36	S001	Ana	Cruz	ana.cruz@cca.edu.ph	D-101	1	Freshmen	IT305	Mr. Santos
37	S003	Cara	Lim	cara.lim@cca.edu.ph	D-102	1	Freshmen	CS204	Mr. Santos
38	S004	Dan	Santos	dan.santos@cca.edu.ph	D-102	3	Junior and Senior	IT101	Mr. Cruz

STUDENT TABLE:

Queries [6]

- Student_Flat_Data
- practice
- stud_dim
- course_dim
- dept_dim
- info_Fact

```
= Table.Distinct(#"Removed Other Columns")
```

	StudentID	First Name	Last Name	Email
1	S002	Ben	Reyes	ben.reyes@cca.edu.ph
2	S004	Dan	Santos	dan.santos@cca.edu.ph
3	S001	Ana	Cruz	ana.cruz@cca.edu.ph
4	S003	Cara	Lim	cara.lim@cca.edu.ph
5	S005	Eva	Flores	eva.flores@cca.edu.ph

COURSE TABLE:

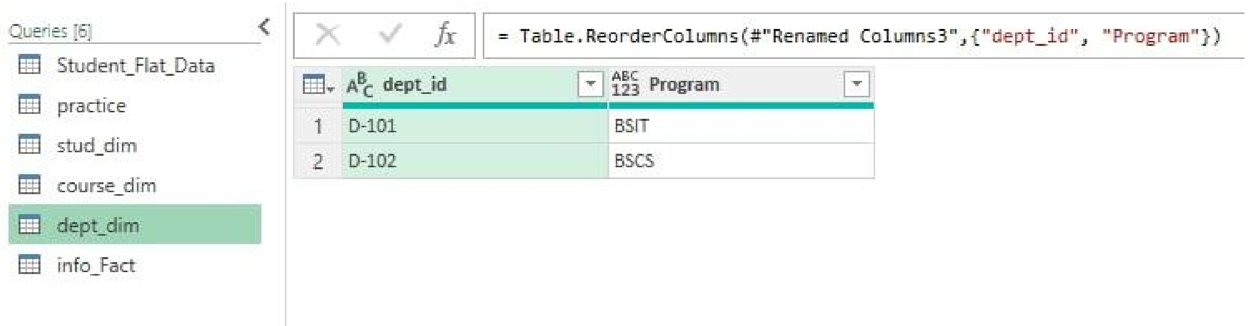
Queries [6]

- Student_Flat_Data
- practice
- stud_dim
- course_dim
- dept_dim
- info_Fact

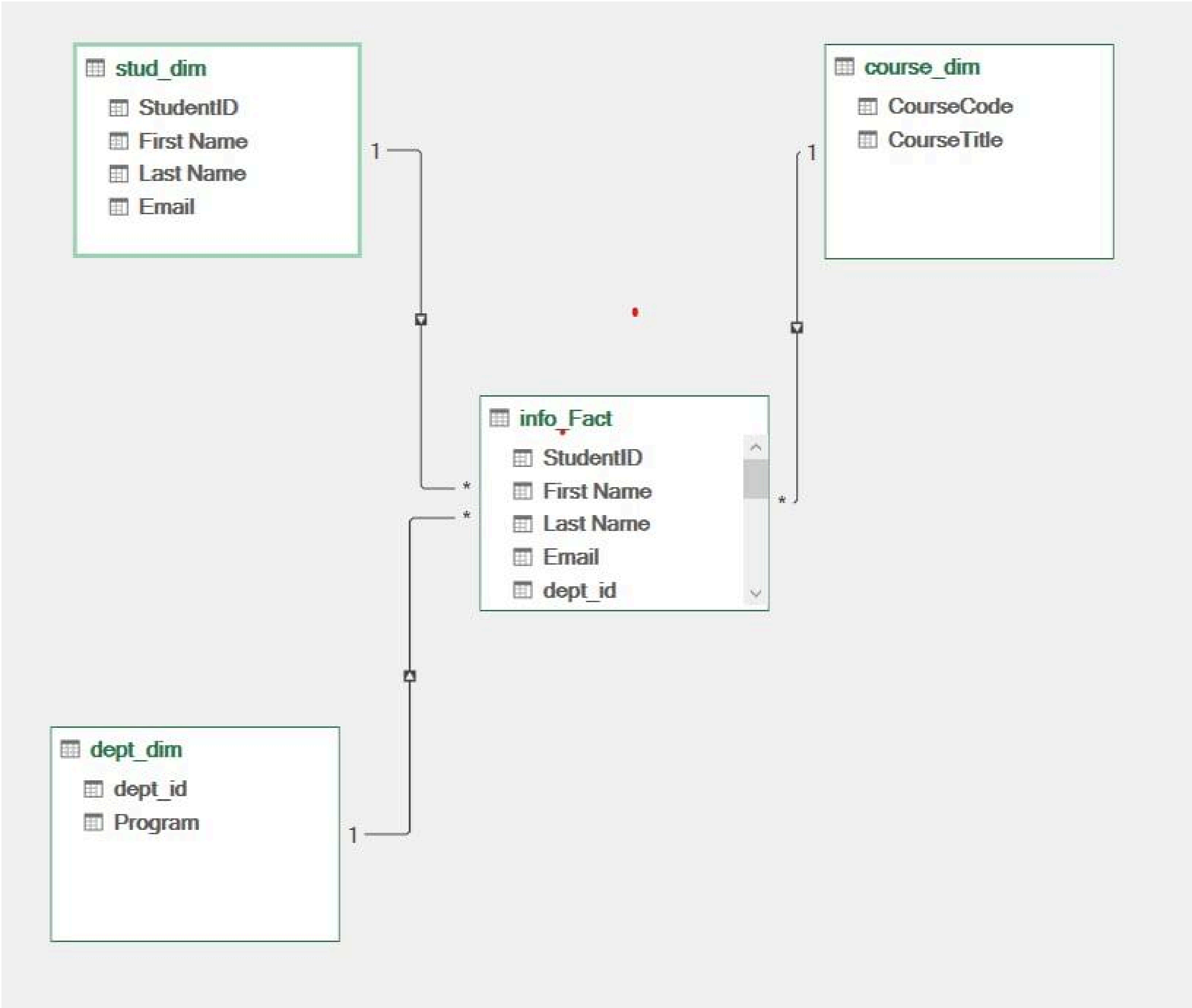
```
= Table.Distinct(#"Removed Other Columns", {"CourseCode"})
```

	CourseCode	CourseTitle
1	CS204	Object-Oriented Programming
2	IT203	Database Systems
3	IT101	Introduction to IT
4	CS102	Programming Fundamentals
5	IT305	Operating Systems

DEPARTMENT TABLE:



DATA MODEL:


stud_dim
StudentID
First Name
Last Name
Email
course_dim
CourseCode
CourseTitle
info_Fact
StudentID
First Name
Last Name
Email
dept_id
dept_dim
dept_id
Program
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